# MD SHANIAN MOED

Electrical & Electronic Engineering | Semiconductor Device Modeling | PhD Research Applicant

Dhaka, Bangladesh • +880 1521-495146 • shanian.eee@aust.edu • shanian.aca@gmail.com
AUST Faculty Profile • LinkedIn • Website

## RESEARCH INTERESTS

Semiconductor TCAD and quantum transport • Nanoscale FETs and GaN HEMTs • Multiphysics FEM and optoelectronic modeling • 2D materials and density functional theory • Lead-free photovoltaics • Plasmonic sensors • Memristors and neuromorphic devices

## EDUCATION

**M.Sc. in Electrical and Electronic Engineering** 2023– ongoing
Ahsanullah University of Science and Technology (AUST), Dhaka

CGPA: **4.00/4.00** (Course)

**Thesis** : 3D finite-element optical–electrical modeling of $Cs_2AgBiBr_6$ double-perovskite solar cells; ongoing extension to coupled thermal analysis. | **Supervisor**: Dr. Md. Tashfiq Bin Kashem

**B.Sc. in Electrical and Electronic Engineering** 2017–2022
Ahsanullah University of Science and Technology (AUST), Dhaka

CGPA: **3.932/4.00**

**Undergraduate thesis**: "Developing a Network-Friendly Charging Scheme for Electric Vehicles". | **Supervisor**: Prof. Dr. Tareq Aziz.

## ACADEMIC APPOINTMENTS

**Lecturer (Grade-I) | Electrical & Electronic Engineering** Dec 2025–Present
Ahsanullah University of Science and Technology (AUST), Dhaka

**Lecturer (Grade-II) | Electrical & Electronic Engineering** Oct 2023–Dec 2025
Ahsanullah University of Science and Technology (AUST), Dhaka

**Lecturer | Electrical & Electronic Engineering** Oct 2022–Oct 2023
Southeast University, Dhaka

## PUBLICATIONS

**Journal article**

**J1.** **Md Shanian Moed**, Adnan Amin Siddiquee, and Md Tashfiq Bin Kashem, "Three-dimensional finite-element-based coupled optical-electrical modeling of $Cs_2AgBiBr_6$ double perovskite solar cells: Device-Level optimization for enhanced efficiency." *Solar Energy Materials and Solar Cells, vol. 304, art. 114441, 2026.* DOI: 10.1016/j.solmat.2026.114441

**IEEE conference papers**

**C1.** **M. S. Moed,** T. Alam, T. A. Khan, S. Abdullah-Al-Nahid, and T. Aziz, "A Genetic Algorithm based Network Friendly Charging Scheme for Electric Vehicles." *2022 International Conference on Green Energy, Computing and Sustainable Technology (GECOST), pp. 423–427, 2022.* DOI: 10.1109/GECOST55694.2022.10010699

**C2.** **M. S. Moed**, S. Mahmud, T. Aziz, S. Abdullah-Al-Nahid, and T. A. Khan, "A consumer-friendly demand side management technique for residential loads supported by two stage Genetic Algorithm-based optimization." *2022 Global Energy Conference (GEC), pp. 229–234, 2022.* DOI: 10.1109/GEC55014.2022.9986733

## SUBMITTED MANUSCRIPTS

**S1.** Reinventing plasmonic biosensing: Novel machine learning optimization of HfO2-assisted architectures for colorectal cancer screening

## SELECTED RESEARCH & MODELING PROJECTS

**Lead-Free Double-Perovskite Photovoltaics | COMSOL 3D FEM** / *Published study; thermal extension ongoing*

Built a coupled Wave Optics (EWFD)–Semiconductor framework for $Cs_2AgBiBr_6$ solar cells, including wavelength-resolved optical generation, drift–diffusion transport, and recombination losses. Investigated transport-layer selection, thickness, doping, defects, temperature, and efficiency. Extending the framework with Heat Transfer in Solids for optical–electrical–thermal analysis.

**AlGaN/GaN MIS-HEMT Device Simulation** / *Ongoing*

Developing a physics-based COMSOL model of T-gate AlGaN/GaN HEMTs on SiC, with attention to polarization, interface charge, doping, gate dielectrics, and source/drain boundary conditions. Target characteristics include output ($I_D$–$V_{DS}$) and transfer ($I_D$–$V_{GS}$) curves.

**CNTFET Ballistic and Quasi-Ballistic Transport** / *Ongoing*

Formulating nanoscale CNTFET transport models and numerical studies using MATLAB and COMSOL. Examining diameter/chirality, gate insulator thickness, electrostatic control, temperature, drain bias, and non-ballistic effects on I–V response and on/off behavior; evaluating quantum-transport extensions beyond classical drift–diffusion.

**Memristor Compact Models and Neuromorphic Devices** / *Ongoing*

Implemented and compared established nonlinear memristor models (Joglekar, Biolek, Prodromakis, and Lehtonen–Laiho) in MATLAB. Studying pinched hysteresis, state evolution, and memristance dynamics, with SPICE/Verilog-A device-model integration as a next step.

**DFT-Based Studies of Two-Dimensional Materials** / *Early-stage research*

Investigating the electronic structure and device-relevant properties of 2D semiconductors using Quantum ESPRESSO and CASTEP. Work includes first-principles structural optimization and MPI-parallel Quantum ESPRESSO calculations, with band-structure and density-of-states analysis as part of ongoing materials screening.

## TECHNICAL SKILLS

**Device & Multiphysics Simulation:** COMSOL Multiphysics (Wave Optics / Electromagnetic Waves, Frequency Domain; Semiconductor; Heat Transfer in Solids); Silvaco TCAD; coupled FEM/PDE and drift–diffusion models.

**DFT & Electronic-Structure Software:** Quantum ESPRESSO (pw.x, SCF / vc-relax, MPI parallel workflows); CASTEP; structural relaxation and electronic-structure analysis (band structure, density of states / projected DOS).

**Circuit & Electronics Tools:** Cadence Virtuoso, MATLAB/Simulink, LTspice, PSpice, ModelSim, Quartus, Proteus; 6T–9T SRAM circuit evaluation.

**Programming & Modeling:** Python, MATLAB, C++, Verilog; parametric sweeps, numerical modeling, data analysis, and scientific visualization.

**Other Engineering Tools:** DIgSILENT PowerFactory, Arduino, FPGA, AutoCAD; Linux-based research environments.

**Methods:** Optical absorption and carrier-generation mapping, semiconductor recombination, device I–V characterization, SPR angular interrogation, genetic-algorithm optimization, compact device models.

## HONORS & SCHOLARSHIPS

- Khan Bahadur Ahsanullah Gold Medal (President Gold Medal), AUST (2025).
- Dean's List of Honor, Ahsanullah University of Science and Technology.
- 50% merit-based tuition waiver during undergraduate study at AUST.
- General Scholarships in SSC and HSC examinations; Talent-pool Scholarships in earlier schooling.

## LANGUAGES & PROFESSIONAL ENGAGEMENT

Bangla (native) • English (professional working proficiency) • IEEE AUST Student Branch (former member) • Departmental academic and project-showcase activities.

## ACADEMIC REFERENCES

**Prof. Dr. Bobby Barua** — Professor, Department of EEE, AUST | bobby@aust.edu

**Dr. Md. Tashfiq Bin Kashem** — Associate Professor, Department of EEE, AUST | tashfiq.eee@aust.edu